We've seen how the state-vector evolves between the time that a system is prepared in a given state and the time that it is brought into contact with an apparatus and measured. If the state-vector were main focus of observational physics, we would say that quantum mechanics is deterministic. But experimental physics is not about measuring the state-vector. It is about measuring observables. Even if we know the state-vector exactly, we don't know the result of any given measurement. Nevertheless, it is fair to say that between observations, the state of a system evolves in a perfectly definite way, according to the time-dependent Schrödinger equation.

But something different happens when an observation is made. An experiment to measure **L** will have an unpredictable outcome, but after the measurement is made, the system is left in an eigenstate of **L**. Which eigenstate? The one corresponding to the outcome of the measurement. But this outcome is unpredictable. So it follows that during an experiment the state of a system jumps unpredictably to an eigenstate of the observable that was measured. This phenomenon is called *the collapse of the wave function.*[4]

To put it another way, suppose the state-vector is

$$\sum_j \alpha_j |\lambda_j\rangle$$

just before the measurement of **L**. Randomly, with probability $|\alpha_j|^2$, the apparatus measures a value λ_j and leaves the system in a single eigenstate of **L**, namely $|\lambda_j$. The entire superposition of states collapses to a single term.

This strange fact—that the system evolves one way between measurements and another way during a measurement—has been a source of contention and confusion for decades. It raises a question: Shouldn't the act of measurement itself be described by the laws of quantum mechanics?

The answer is yes. The laws of quantum mechanics are not suspended during measurement. However, to examine the measurement process itself as a quantum mechanical evolution, we must consider the entire experimental setup, including the apparatus, as part of a single quantum system. We'll